Errata for: **"Probing the Universe: A Geometrical View for Observers of Spacetime Physics"**

Robin W. Tucker and Timothy J. Walton (Springer: 2026)

A list of typographical errors and correction for the book will be kept here. Corrections are emphasised in **red** text.

Please email TWalton@lancashire.ac.uk if you find any typographical errors not already listed.

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page 16: The third displayed equation on this page should read:

$$\boldsymbol{v}_P(t) \;=\; -R_0 \sin(\,\Phi(t)\,)\,\dot{\boldsymbol{\Phi}}(t)\,\boldsymbol{i}_1 + R_0 \cos(\,\Phi(t)\,)\,\dot{\boldsymbol{\Phi}}(t)\,\boldsymbol{i}_2$$

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page 74: The displayed equation below equation (5.36) should read:

$$\begin{aligned} 0 \diamond \boldsymbol{\xi} \;&=\; \boldsymbol{\xi} \\ \tau_1 \diamond (\tau_2 \diamond \boldsymbol{\xi}) \;&=\; (\tau_1 \boxtimes \tau_2) \diamond \boldsymbol{\xi} \end{aligned}$$

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page 177: The line immediately below the first displayed equation should read:

"In the $\mathbf{g}_{\mathbb{R}^3}$-orthonormal cobasis for…"

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page 248: The line immediately below the penultimate displayed equation on this page should read:

"Since the g-trace of g is $\mathrm{Tr}^{(\mathrm{g})}(\mathrm{g}) = 4$, taking…"

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page 285: The line directly above the only displayed equation on this page should read:

"Since $\Pi^{\perp}_{Q}(W') = \boldsymbol{W}$, both the spacelike…"

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page 290: The sentence directly preceding equation (15.45) should read:

"Recalling $W \equiv \boldsymbol{\Pi}^{\perp}_{\mathbf{Q}}(W')$ where Q is geodesic with $\mathrm{g}(Q,Q) = -L_0^2$ and W' is Lie-parallel with respect to Q, the vector fields Q, W **satisfy** the conditions…"

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page 376: The last displayed equation on this page should read:

$$\begin{aligned}\mathcal{G}_c &= \mathcal{R}\star e_c - 2(i_a P_c)\star e^a + \mathbf{2}(i_c i_a i_b\,\mathfrak{D}T^b)\star e^a \\ &= \mathcal{R}\star e_c - 2\star P_c - 2i_a(i_c i_b\,\mathfrak{D}T^b)\star e^a\end{aligned}$$

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page 380: The line immediately below the first displayed equation should read:

"...is a type-preserving *derivation* on the algebra of exterior forms on $\mathcal{M}$ **(see appendix H for further details)** that commutes with the exterior derivative..."

There is also a typographical error here: the 8th line below equation (20.8) should read

"...equality modulo d will below be **denoted** by..."

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page 382: The line immediately below equation (20.13) should read:

"...of the interior operator $i_c \equiv i_{\boldsymbol{X_c}}$,..."

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page 383: The top displayed equation on this page should read:

$$\psi_2 = \dot{e}^{\boldsymbol{a}}\wedge e^b\wedge\star R_{ab}.$$

The displayed equation between (20.17) and (20.18) should read:

$$\begin{aligned}\mathfrak{D}\star e^{ab} &= \frac{1}{2}\epsilon^{ab}{}_{cd}\,\mathfrak{D}e^{cd} = \frac{1}{2}\epsilon^{ab}{}_{cd}\left(T^c\wedge e^d - e^c\wedge T^d\right) \\ &= \epsilon^{ab}{}_{cd}\,T^c\wedge e^d = T_c\wedge\star e^{abc}\end{aligned}$$

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page 384: The line of text above equation (20.22) should read:

"In this case the 3-forms $\{\mathcal{G}_c^{(\mathrm{EH})} \equiv -\frac{\mathbf{1}}{\mathbf{2}}\mathcal{G}_c\}$ define..."

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page 385: Equation (20.24) should read:

$$\begin{aligned}\Lambda_{\mathrm{TOTAL}}(e,\omega,A,\phi,\overline{\phi}) \equiv\ & \frac{\mathbf{1}}{\mathbf{2}\boldsymbol{\kappa}}R_{ab}(\omega)\wedge\star e^{ab} - \frac{\mathbf{1}}{\mathbf{2}}\kappa_A dA\wedge\star dA \\ & + \frac{\mathbf{1}}{\mathbf{2}}\kappa_1\mathrm{Re}\left(\overline{\mathfrak{D}_{\mathrm{U}(1)}\phi}\wedge\star\mathfrak{D}_{\mathrm{U}(1)}\phi\right) \\ & + \frac{\mathbf{1}}{\mathbf{2}}\kappa_2\,\phi\,\overline{\phi}\star 1 + \kappa_3\,R_{ab}(\omega)\wedge R^{ab}(\omega)\end{aligned} \tag{20.24}$$

The change in minus sign here could be absorbed into the coupling constant κ_A but is changed to be consistent with later sections.

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page 386: Equation (20.30) should read:

$$\mathcal{I}_{\text{EH}}[e,\omega] \;\equiv\; \frac{\mathbf{1}}{\mathbf{2}\kappa}\int_{\mathcal{M}} R_{ab}(\omega)\wedge\star(\,e^a\wedge e^b\,), \tag{20.30}$$

and the displayed equation below (20.30) becomes:

$$\mathcal{G}_c^{(\text{EH})} \;\equiv\; -\frac{\mathbf{1}}{\mathbf{2}}\,R_{ab}\wedge i_c\star(\,e^a\wedge e^{\boldsymbol{b}}\,) \;=\; 0, \quad a,b,c=0,1,2,3,\; e^a(X_c) \;=\; \delta^a_c$$

$$T^c \;=\; 0.$$

page 388: Equation (20.31) should read:

$$\texttt{Ein} \;\equiv\; (\star^{-1}\mathcal{G}_c^{(\text{EH})})\otimes e^c \;=\; \kappa\,(\star^{-1}\tau_c^{(\text{SOURCE})})\otimes e^c. \tag{20.31}$$

page 392: The first equation of section 20.4 should read:

$$\mathcal{I}_{\text{EH}}[e,\omega] \;\equiv\; \frac{\mathbf{1}}{\mathbf{2}\boldsymbol{\kappa}}\int_{\mathcal{M}} R_{ab}(\omega)\wedge\star(\,e^a\wedge e^b\,), \tag{20.30}$$

and the second equation becomes:

$$\mathcal{I}_{\text{SOURCE}}[e,A,\phi,\overline{\phi}] \;\equiv\; -\frac{\mathbf{1}}{\mathbf{2}}\kappa_A\int_{\mathcal{M}}\Lambda_2(e,A)+\frac{\mathbf{1}}{\mathbf{2}}\kappa_1\int_{\mathcal{M}}\Lambda_1(e,A,\phi,\overline{\phi})$$
$$+\frac{\mathbf{1}}{\mathbf{2}}\mu^2\int_{\mathcal{M}}\Lambda_3(e,\phi,\overline{\phi}) \qquad (\kappa_1,\mu\in\mathbb{R})$$

page 393: The third displayed equation on this page should read:

$$\widehat{\boldsymbol{\delta}}\boldsymbol{\Lambda_3}(\boldsymbol{e},\boldsymbol{\phi},\overline{\boldsymbol{\phi}}) \;=\; \text{Re}\left(\dot{\phi}\wedge\star\overline{\phi}+\phi\wedge\star\dot{\overline{\phi}}+\dot{e}^c\wedge\Big[\,\phi\wedge i_c\star\overline{\phi}-i_c\overline{\phi}\wedge\star\phi\Big]\right).$$

The fourth displayed equation should be:

$$\widehat{\boldsymbol{\delta}}\boldsymbol{\Lambda_3}(\boldsymbol{e},\boldsymbol{\phi},\overline{\boldsymbol{\phi}}) \;=\; \text{Re}\left(\dot{\phi}\,\overline{\phi}+\dot{\overline{\phi}}\,\phi\right)\star 1+\dot{e}^c\wedge\Big[\,\phi\,\overline{\phi}\star e_c\Big],$$

the fifth displayed equation should read:

$$\begin{aligned}\widehat{\boldsymbol{\delta}}\boldsymbol{\Lambda_2}(\boldsymbol{e},\boldsymbol{A}) \;&=\; d\widehat{\delta}A\wedge\star dA+dA\wedge\star\widehat{\delta}dA+\dot{e}^c\wedge[\,dA\wedge i_c\star dA-i_c dA\wedge\star dA\,]\\ &=\; d\dot{A}\wedge\star F+F\wedge\star d\dot{A}+\dot{e}^c\wedge[\,F\wedge i_c\star F-i_c F\wedge\star F\,]\end{aligned}$$

and the last displayed equation on the page should be:

$$\widehat{\boldsymbol{\delta}}\boldsymbol{\Lambda_2}(\boldsymbol{e},\boldsymbol{A}) \;\simeq\; 2\,\dot{A}\wedge d\star F+\dot{e}^c\wedge[\,F\wedge i_c\star F-i_c F\wedge\star F\,].$$

page 394: The first displayed equation on this page should read:

$$\widehat{\delta}\boldsymbol{\Lambda_1}(\boldsymbol{e},\boldsymbol{A},\boldsymbol{\phi},\overline{\boldsymbol{\phi}}) \;=\; \mathrm{Re}\,\Big(\widehat{\delta}\mathfrak{D}_{\mathrm{U}(1)}\overline{\phi}\wedge\star\mathfrak{D}_{\mathrm{U}(1)}\phi+\widehat{\delta}\mathfrak{D}_{\mathrm{U}(1)}\phi\wedge\star\mathfrak{D}_{\mathrm{U}(1)}\overline{\phi}$$
$$+\,\dot{e}^c\wedge\big[-\mathfrak{D}_{\mathrm{U}(1)}\overline{\phi}\wedge i_c\star\mathfrak{D}_{\mathrm{U}(1)}\phi-i_c\mathfrak{D}_{\mathrm{U}(1)}\phi\wedge\star\mathfrak{D}_{\mathrm{U}(1)}\overline{\phi}\big]\,\Big)\,.$$

The second displayed equation on this page should read:

$$\widehat{\delta}\mathfrak{D}_{\mathrm{U}(1)}\phi \;=\; d\dot{\phi}+i\hat{s}\,A\,\dot{\phi}+i\hat{s}\,\dot{A}\,\phi \;=\; \mathfrak{D}_{\mathrm{U}(1)}\dot{\phi}+i\hat{s}\,\dot{A}\,\phi$$
$$\widehat{\delta}\,\mathfrak{D}_{\mathrm{U}(1)}\overline{\phi} \;=\; d\dot{\overline{\phi}}-i\hat{s}\,A\,\dot{\overline{\phi}}-i\hat{s}\,\dot{A}\,\overline{\phi} \;=\; \overline{\mathfrak{D}}_{\mathrm{U}(1)}\,\dot{\overline{\phi}}-i\hat{s}\,\dot{A}\,\overline{\phi}.$$

The last equation on the page should be:

$$\widehat{\delta}\boldsymbol{\Lambda_1}(\boldsymbol{e},\boldsymbol{A},\boldsymbol{\phi},\overline{\boldsymbol{\phi}}) \;=\; \mathrm{Re}\left\{\mathfrak{D}_{\mathrm{U}(1)}\dot{\overline{\phi}}\wedge\star\mathfrak{D}_{\mathrm{U}(1)}\phi+\mathfrak{D}_{\mathrm{U}(1)}\dot{\phi}\wedge\star\mathfrak{D}_{\mathrm{U}(1)}\overline{\phi}\right\}$$
$$-\;2\hat{s}\,\dot{A}\;\wedge\mathrm{Im}\left(\phi\wedge\star\overline{\mathfrak{D}_{\mathrm{U}(1)}\phi}\right)$$
$$-\;\dot{e}^c\wedge\mathrm{Re}\left(\mathfrak{D}_{\mathrm{U}(1)}\phi\wedge i_c\star\mathfrak{D}_{\mathrm{U}(1)}\overline{\phi}-i_c\mathfrak{D}_{\mathrm{U}(1)}\overline{\phi}\wedge\star\mathfrak{D}_{\mathrm{U}(1)}\phi\right)\,.$$

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page 395: The fourth displayed equation on this page should read:

$$\mathrm{Re}\left\{\mathfrak{D}_{\mathrm{U}(1)}\dot{\phi}\wedge\star\overline{\mathfrak{D}_{\mathrm{U}(1)}\phi}+\mathfrak{D}_{\mathrm{U}(1)}\dot{\overline{\phi}}\wedge\star\mathfrak{D}_{\mathrm{U}(1)}\phi\right\}$$
$$\simeq\;\mathrm{Re}\left\{-\dot{\phi}\wedge\mathfrak{D}_{\mathrm{U}(1)}\star\mathfrak{D}_{\mathrm{U}(1)}\overline{\phi}-\dot{\overline{\phi}}\wedge\mathfrak{D}_{\mathrm{U}(1)}\star\mathfrak{D}_{\mathrm{U}(1)}\phi\right\}.$$

The fifth displayed equation on this page should read:

$$\widehat{\delta}\boldsymbol{\Lambda_1}(\boldsymbol{e},\boldsymbol{A},\boldsymbol{\phi},\overline{\boldsymbol{\phi}}) \;\simeq\; \mathrm{Re}\left\{-\dot{\phi}\wedge\mathfrak{D}_{\mathrm{U}(1)}\star\mathfrak{D}_{\mathrm{U}(1)}\overline{\phi}-\dot{\overline{\phi}}\wedge\mathfrak{D}_{\mathrm{U}(1)}\star\mathfrak{D}_{\mathrm{U}(1)}\phi\right\}$$
$$-\;2\hat{s}\,\dot{A}\wedge\mathrm{Im}\left(\phi\wedge\star\mathfrak{D}_{\mathrm{U}(1)}\overline{\phi}\right)$$
$$-\;\dot{e}^c\wedge\mathrm{Re}\left(\mathfrak{D}_{\mathrm{U}(1)}\phi\wedge i_c\star\overline{\mathfrak{D}_{\mathrm{U}(1)}\phi}-i_c\overline{\mathfrak{D}_{\mathrm{U}(1)}\phi}\wedge\star\mathfrak{D}_{\mathrm{U}(1)}\phi\right)\,.$$

The displayed equation immediately above equation (20.37) should read:

$$\widehat{\widehat{\delta}}\left(\mathcal{I}_{\mathrm{EH}}[e,\omega]+\mathcal{I}_{\mathrm{SOURCE}}[e,A,\phi,\overline{\phi}]\right)\;\simeq\;0,$$

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page 396: Equation (20.38) should read:

$$\kappa_A \, d\star F + \kappa_1 \, \hat{s} \, \mathrm{Im}\left(\phi \star \overline{\mathfrak{D}_{\mathrm{U}(1)}\phi}\right) \;=\; 0 \tag{0.1}$$

and the equation immediately following it becomes:

$$\mathcal{J} \;\equiv\; -\,\frac{\epsilon_0\,\kappa_1\,\hat{s}}{\kappa_A}\,\mathrm{Im}\left(\phi \star \overline{\mathfrak{D}_{\mathrm{U}(1)}\phi}\right).$$

Equation (20.39) should read:

$$\frac{\mathcal{G}_c^{(\mathrm{EH})}}{\kappa} \;=\; \tau_c \tag{20.39}$$

and the text immediately after this displayed equation becomes:

"where $\mathcal{G}_c^{(\mathrm{EH})} \equiv -\frac{1}{2} R_{ab} \wedge i_c \star e^{ab} \ldots$".

The displayed equation after (20.39) should read:

$$\begin{aligned}\tau_c \;=\; & \frac{\mu^2}{2}\,\phi\,\overline{\phi}\star e_c + \frac{\kappa_A}{2}\,\left(\boldsymbol{i_c F \wedge \star F - F \wedge i_c \star F}\right) \\ & - \frac{\boldsymbol{\kappa_1}}{\boldsymbol{2}}\,\mathrm{Re}\left(\mathfrak{D}_{\mathrm{U}(1)}\phi \wedge i_c \star \mathfrak{D}_{\mathrm{U}(1)}\overline{\phi} - i_c\overline{\mathfrak{D}_{\mathrm{U}(1)}\phi} \wedge \star\mathfrak{D}_{\mathrm{U}(1)}\phi\right) \qquad (c = 0, 1, 2, 3).\end{aligned}$$

The text in the first paragraph immediately below the last displayed equation of this page should be:

"Note $[\tau_c] = \mathbf{M}[c_0^2]$. Since the $\dot{\omega}_{ab}$ partial variations in $\mathcal{I}_{\mathrm{EH}}[e,\omega]$ *and* $\mathcal{I}_{\mathrm{SOURCE}}[e, A, \phi, \overline{\phi}]$ yield a set of zero-torsion connection 1-forms (see section above), if these are assumed to be metric-compatible with respect to the metric tensor $\mathrm{g} = \eta_{ab}\, e^a \otimes e^b$ then (20.37), (20.38), (20.20) constitute the necessary conditions for the functional $\mathcal{I}_{\mathbf{EH}}[e,\omega] + \mathcal{I}_{\mathbf{SOURCE}}[e, A, \phi, \overline{\phi}]$ to be extremal and (20.39) yields a *Levi-Civita Einstein tensor field equation with sources*. "

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page 402: The penultimate displayed equation on this page should read:

$$\mathrm{T}^{(\mathrm{MAX})}(X_0, X_0) \;=\; (\star\tau_c^{(\mathrm{MAX})})(X_0)\,\boldsymbol{\delta_0^c} \;=\; i_0(\star\tau_0^{(\mathrm{MAX})})$$

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page 405: The second displayed equation in section 21.1 should read:

$$(\nabla\cdot\Xi)(\boldsymbol{Y}) \;\equiv\; (\nabla_{X_a}\Xi)(X^a, Y) \qquad \text{for all } Y \in \Gamma T\mathcal{U}_{\mathcal{M}}.$$

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page 414: Equation (22.8) should read:

$$\mathcal{G}_a^{(\mathrm{EH})} \;=\; \kappa\, \tau_a^{(\mathrm{SOURCE})}, \qquad a = 0, 1, 2, 3. \tag{22.8}$$

The text immediately following this line should then read:

"Recall that it follows from $\left[\mathcal{G}_a^{(\mathrm{EH})}\right]$ and ...".

Equation (22.9) should read:

$$\mathcal{G}_a^{(\mathrm{EH})} \;=\; \kappa \sum_N \tau_a^N \;\equiv\; \kappa\, \tau_a^{(\mathrm{SOURCE})}. \tag{22.9}$$

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page 488: The first displayed equation on this page should read:

$$\begin{aligned}
\frac{1}{L_0} \mathrm{g}_{\mathrm{PG}}(\dot{C}, X_0)\Big|_{C(\tau)} &= -\dot{T}(\tau) \\
\frac{1}{L_0^2} \mathrm{g}_{\mathrm{PG}}(\dot{C}, \dot{C})\Big|_{C(\tau)} &= -\left(1 - \frac{\Sigma}{R(\tau)}\right) \dot{T}(\tau)^2 + 2\sqrt{\frac{\Sigma}{R(\tau)}}\, \dot{T}(\tau)\dot{R}(\tau) + \dot{R}(\tau)^2 \\
&\qquad + R(\tau)^2 \left(\dot{\Theta}(\tau)^2 + \sin^2(\,\boldsymbol{\Theta}(\tau)\,)\dot{\Phi}(\tau)^2\right).
\end{aligned}$$

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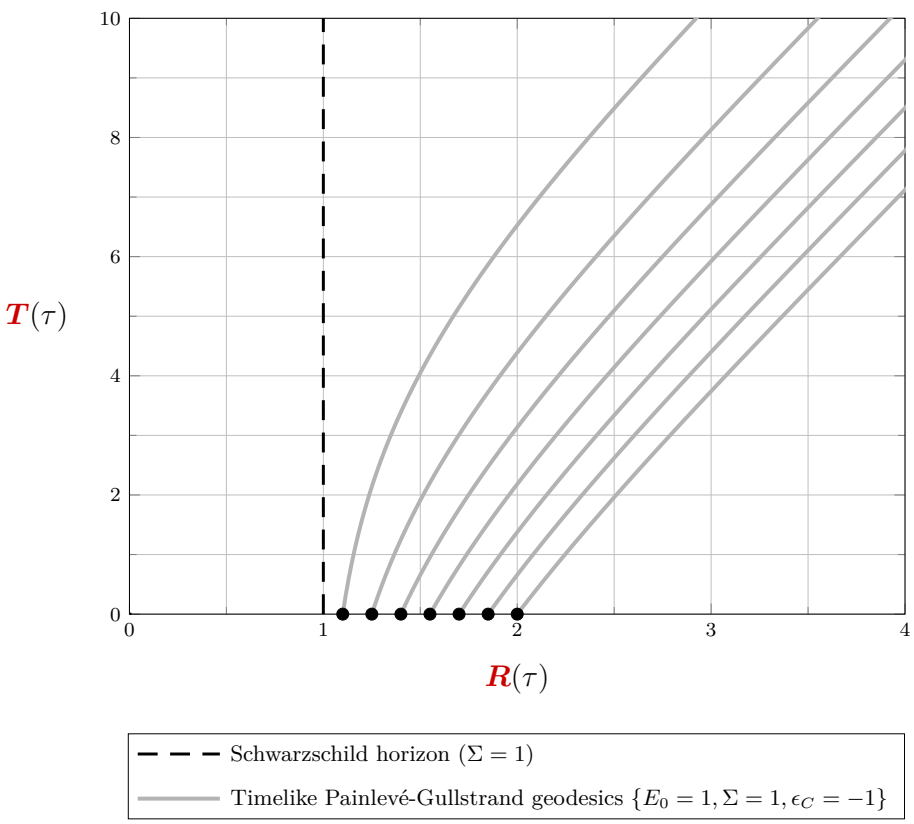
page 491: The axis labels $(\,T(\tau), R(\tau)\,)$ in Fig. 30.1 are the wrong way round and need to be swapped. The figure should be as follows:



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page 573: The penultimate displayed equation on this page should read:

$$f(r^*) \equiv r^* + \frac{(\Sigma_1^2 - 2\boldsymbol{\Sigma_2^2})}{\sqrt{4\Sigma_2^2 - \Sigma_1^2}}\arctan\left(\frac{2r^* - \Sigma_1}{\sqrt{4\Sigma_2^2 - \Sigma_1^2}}\right) + \frac{\Sigma_1}{2}\ln\left((r^*)^2 - \Sigma_1 r^* + \Sigma_2^2\right).$$

page 574: The first displayed equation on this page has an extra dr^* in the first term that needs removing. It should read:

$$\begin{cases} \Psi^* dt = d(\Psi^* t) = d(t^* + f(r^*)) = dt^* + \dfrac{dr^*}{1 - \frac{\Sigma_1}{r^*} + \frac{\Sigma_2^2}{(r^*)^2}} \\ \Psi^* dr = d(\Psi^* r) = dr^* \\ \Psi^* d\theta = d(\Psi^* \theta) = d\theta^* \\ \Psi^* d\phi = d(\Psi^* \phi) = d(\phi^*) = d\phi^* \end{cases}$$

The Reissner-Nordström -star metric (30.180) should then read:

$$\frac{\mathrm{g}_{\mathrm{RN}^*}}{L_0^2} \equiv \frac{\Psi^* \mathrm{g}_{\mathrm{RN}}^{(t,r,\theta,\phi)}}{L_0^2} = -\left(1 - \frac{\Sigma_1}{r^*} + \frac{\Sigma_2^2}{(r^*)^2}\right) dt^* \otimes dt^* - dt^* \otimes dr^* - dr^* \otimes dt^*$$
$$+ (r^*)^2 \left(d\theta^* \otimes d\theta^* + \sin^2(\theta^*)\, d\phi^* \otimes d\phi^*\right). \qquad (30.180)$$

page 659: Equation (34.14) for the Bopp-Podolsky field equations should read:

$$d \star \mathcal{F} + \lambda_0^2\, d\delta d \star \mathcal{F} = 0 \qquad (34.14)$$

The change in sign here ensures the static, spherically symmetric solution is finite as $r \to 0$. This correction contributes to a number of changes over the next few pages.

page 660: Equation (34.16) for the Bopp-Podolsky field equations should read:

$$d \star \mathcal{F} + \lambda_0^2\, d\delta d \star \mathcal{F} = d\delta d\alpha \wedge \Pi + \lambda_0^2\, d\delta d\delta d\alpha \wedge \Pi$$
$$= d\left(\delta d\alpha + \lambda_0^2\, \delta d\delta d\alpha\right) \wedge \Pi = 0. \qquad (34.16)$$

The following equation also needs correcting:

$$\Box\alpha + \lambda_0^2 \Box^2 \alpha = 0, \qquad \Box\alpha \neq 0, \qquad (34.17)$$

as does the final displayed equation of the page which should read:

$$\Box\alpha + \frac{1}{\lambda_0^2}\alpha = 0, \qquad \lambda_0 \neq 0.$$